

Point

From Number Theory to Algebraic Geometry and Logic

Session S3 — Fields, Extensions, and Algebraic Closure

Complete lecture notes with proofs, examples, and exercises

Abstract

This session builds the *arenas* in which the course's points live: fields and their extensions. A field is where polynomials have (or fail to have) roots — i.e. where algebraic points exist; an extension L/K measures how many new points appear; the minimal polynomial is the equation that cuts a point out; and the algebraic closure $\bar{K}$ is the space of *all* algebraic points over K , whose symmetry group $\text{Gal}(\bar{K}/K)$ — for $K = \mathbb{Q}$ the group $G_{\mathbb{Q}}$ — is the horizon of the entire course (S16, S35, S45). Everything is proved from the ring theory of S1: Krull's theorem (via Zorn) is exactly what makes algebraic closures exist.

Session plan (150 minutes)

Time	Block
0:00–0:10	Orientation: fields as arenas of points; where S3 feeds (S12–S16, S20, S45)
0:10–0:35	§1 Fields, characteristic, prime subfields, finite fields
0:35–1:05	§2 Extensions, degree, tower law, minimal polynomials, simple extensions
1:05–1:15	<i>Break</i>
1:15–1:45	§3 Algebraic closure: existence via Zorn (S1), uniqueness, $\bar{\mathbb{Q}}$ vs $\mathbb{C}$
1:45–2:10	§4 Preview: $\text{Gal}(L/K)$, the group $G_{\mathbb{Q}}$, profinite horizon (S16, S34)
2:10–2:30	Exercises / wrap-up: algebraic closure = space of all algebraic points

Conventions

All fields are commutative with $1 \neq 0$. An *extension* L/K means $K \subseteq L$ with the same operations; we write $[L : K]$ for its degree. $K[x]$ denotes the polynomial ring and $K(x)$ the field of rational functions. By S1, every nonzero ring has a maximal ideal (Krull, via Zorn) — this is used essentially in §3.

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1 Fields, Characteristic, and Finite Fields

Definition 1.1 (Field). A **field** is a commutative ring K with $1 \neq 0$ in which every $a \neq 0$ has a multiplicative inverse a^{-1} . Equivalently: the only ideals of K are 0 and K (S1).

Example 1.2 (Fields old and new). • $\mathbb{Q}, \mathbb{R}, \mathbb{C}$; the p -adic fields $\mathbb{Q}_p$ (S18); function fields $k(x)$ (S21).

- $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ for prime p (a field because (p) is maximal, S1).
- $K = \mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ — a first “point-adding” extension (§2).
- Non-examples: $\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}$ for composite n , $k[x]$ (ideals $(x), (x^2)$ are proper and nonzero).

Definition 1.3 (Characteristic). The **characteristic** $\text{char}(K)$ is the smallest $n \geq 1$ with $\underbrace{1 + \cdots + 1}_n = 0$; if no such n exists, $\text{char}(K) = 0$.

Proposition 1.4 (Characteristic is 0 or prime). $\text{char}(K) \in \{0\} \cup \{\text{primes}\}$.

Proof. If $\text{char}(K) = n = ab$ with $1 < a, b < n$, then $0 = n \cdot 1 = (a \cdot 1)(b \cdot 1)$; a field has no zero-divisors, so $a \cdot 1 = 0$ or $b \cdot 1 = 0$, contradicting minimality of n . $\square$

Proposition 1.5 (Prime subfield). *The intersection of all subfields of K is a subfield, the **prime subfield**, isomorphic to $\mathbb{Q}$ if $\text{char}(K) = 0$ and to $\mathbb{F}_p$ if $\text{char}(K) = p$.*

Proof. The unique ring map $\varphi : \mathbb{Z} \rightarrow K$, $n \mapsto n \cdot 1$, has kernel $(\text{char } K)$ by definition; if $\text{char } K = p$ then φ descends to an embedding $\mathbb{F}_p \hookrightarrow K$; if $\text{char } K = 0$ then φ extends to an embedding $\mathbb{Q} \hookrightarrow K$ (universal property of localization, S1/S2). Any subfield contains 1, hence the image; so the image is the smallest subfield. $\square$

Proposition 1.6 (Order of a finite field). *If $|K| < \infty$ then $|K| = p^n$ where $p = \text{char}(K)$ and $n = [K : \mathbb{F}_p]$.*

Proof. $\text{char}(K) = p > 0$ else $\mathbb{Q} \subseteq K$ would be infinite (Proposition 1.5). Then K is an $\mathbb{F}_p$ -vector space; finiteness forces finite dimension n , and counting gives $|K| = p^n$. $\square$

Proposition 1.7 (Finite multiplicative subgroups are cyclic). *Every finite subgroup $G \subseteq K^\times$ of a field is cyclic. In particular $(\mathbb{Z}/p)^\times$ and $\mathbb{F}_q^\times$ are cyclic (the primitive-root engine of S5).*

Proof. Let $m = |G|$ and let e be the exponent of G (lcm of element orders). Every $g \in G$ is a root of $x^e - 1$, which has at most e roots in K ; since G supplies m distinct roots, $m \leq e$. Always $e \mid m \dots$ more precisely $e \leq m$ and $m \mid e$? Standard finish: $e \leq m$ by the root bound and $m \leq e$ since some element realizes... cleanly: the root bound gives $m \leq e$; Lagrange (P1) gives $\text{ord}(g) \mid m$ for all g , hence $e \mid m$, so $e = m$. An element of order $e = m$ exists (else the exponent of a proper...) — take g with $\text{ord}(g) = e$: exists because e is an lcm of orders and in an abelian group orders compose: if $\text{ord}(a) = r$, $\text{ord}(b) = s$ then some element has order $\text{lcm}(r, s)$ (write $r = r_1 r_2$ coprime pieces); hence some g has order $e = m$ and $G = \langle g \rangle$. $\square$

Remark 1.8 (What is postponed to S12). Existence and uniqueness of a field $\mathbb{F}_{p^n}$ for each p, n (as the splitting field of $x^{p^n} - x$), and the lattice of subfields $\mathbb{F}_{p^d}$, $d \mid n$, are proved in S12 together with the Frobenius automorphism — the first canonical “point-moving” symmetry of the course.

2 Extensions, Degree, and Minimal Polynomials

Definition 2.1 (Extension and degree). An **extension** L/K is a field L containing K . Then L is a K -vector space; its dimension is the **degree** $[L : K]$. The extension is **finite** if $[L : K] < \infty$.

Example 2.2. • $[\mathbb{C} : \mathbb{R}] = 2$ with basis $\{1, i\}$; $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$ with basis $\{1, \sqrt{2}\}$.

- $[\mathbb{R} : \mathbb{Q}] = \infty$ (else $\mathbb{R}$ would be countable — it is not).
- $[K(x) : K] = \infty$: $1, x, x^2, \dots$ are linearly independent.

Proposition 2.3 (Tower law). *If $K \subseteq M \subseteq L$ then $[L : K] = [L : M][M : K]$ (with the convention $\infty \cdot n = \infty$).*

Proof. Let $\{m_i\}_{i \in I}$ be a K -basis of M and $\{l_j\}_{j \in J}$ an M -basis of L . Then $\{m_i l_j\}_{(i,j) \in I \times J}$ is a K -basis of L : spanning — any $x = \sum_j a_j l_j$ with $a_j = \sum_i c_{ij} m_i$; independence — if $\sum_{i,j} c_{ij} m_i l_j = 0$ then $\sum_i c_{ij} m_i = 0$ for each j (independence of the l_j over M), hence all $c_{ij} = 0$. Counting gives the formula. $\square$

Definition 2.4 (Algebraic and transcendental elements). $\alpha \in L$ is **algebraic over K** if $f(\alpha) = 0$ for some nonzero $f \in K[x]$; otherwise **transcendental**. L/K is **algebraic** if every $\alpha \in L$ is algebraic.

Proposition 2.5 (Minimal polynomial and simple extensions). *Let $\alpha \in L$. The evaluation map $\text{ev}_\alpha : K[x] \rightarrow L$, $f \mapsto f(\alpha)$, is a ring homomorphism with kernel an ideal of the PID $K[x]$.*

- (a) *If $\ker = 0$ then α is transcendental and $K(\alpha) \cong K(x)$, $[K(\alpha) : K] = \infty$.*
- (b) *If $\ker \neq 0$ then $\ker = (m_\alpha)$ for a unique monic irreducible m_α , the **minimal polynomial** of α ; then $K(\alpha) = K[\alpha] \cong K[x]/(m_\alpha)$ and $[K(\alpha) : K] = \deg m_\alpha$, with basis $1, \alpha, \dots, \alpha^{d-1}$, $d = \deg m_\alpha$.*

Proof. $K[x]$ is a PID (S1: Euclidean domain). In case (b), $\ker$ is a nonzero prime ideal (kernel of a map into a field), hence maximal in a PID, so $K[x]/(m_\alpha)$ is a field containing K and α , hence equal to $K(\alpha)$; the cosets of $1, x, \dots, x^{d-1}$ give the claimed basis by Euclidean division. Uniqueness of the monic generator is immediate. $\square$

Example 2.6. • $m_{\sqrt{2}, \mathbb{Q}} = x^2 - 2$; $m_{\sqrt[3]{2}, \mathbb{Q}} = x^3 - 2$ (irreducible by Eisenstein, or degree-3 root test); $m_{i, \mathbb{R}} = x^2 + 1$.

- $\alpha = \sqrt{2} + \sqrt{3}$: $\alpha^2 = 5 + 2\sqrt{6}$, $(\alpha^2 - 5)^2 = 24$, so $m_{\alpha, \mathbb{Q}} = x^4 - 10x^2 + 1$; hence $[\mathbb{Q}(\sqrt{2} + \sqrt{3}) : \mathbb{Q}] = 4$, and since $\mathbb{Q}(\sqrt{2} + \sqrt{3}) \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{3})$ with $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] \leq 4$ (tower law), equality holds: $\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2} + \sqrt{3})$ — a primitive element (preview of S12).
- π is transcendental over $\mathbb{Q}$ (Lindemann–Weierstrass; out of scope) so $[\mathbb{Q}(\pi) : \mathbb{Q}] = \infty$.

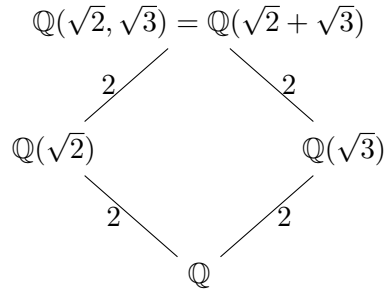
Proposition 2.7 (Finite $\Rightarrow$ algebraic; f.g. algebraic $\Rightarrow$ finite). (a) *If $[L : K] < \infty$ then L/K is algebraic.*

- (b) *If $L = K(\alpha_1, \dots, \alpha_r)$ with all α_i algebraic over K , then $[L : K] < \infty$.*

Proof. (a) For $\alpha \in L$, the vectors $1, \alpha, \dots, \alpha^{[L:K]}$ are linearly dependent, giving a nonzero annihilating polynomial. (b) Induction on r with the tower law: each step $K(\alpha_1, \dots, \alpha_i)$ over $K(\alpha_1, \dots, \alpha_{i-1})$ is finite because α_i is algebraic over K , hence over the larger field (its minimal polynomial over K still annihilates it), so Proposition 2.5(b) bounds the degree. $\square$

Corollary 2.8 (Algebraic elements form a field). *For $K \subseteq L$, the set $\{\alpha \in L : \alpha \text{ algebraic over } K\}$ is a subfield of L (the **algebraic closure of K in L**).*

Proof. For algebraic $\alpha, \beta \neq 0$, the extension $K(\alpha, \beta)$ is finite (Proposition 2.7(b)), hence algebraic (part (a)); and $\alpha \pm \beta, \alpha\beta, \alpha^{-1} \in K(\alpha, \beta)$, so all are algebraic. $\square$



degrees multiply along towers (Proposition 2.3): $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4$

Figure 1: A tower of simple extensions: each step adds one algebraic point, and degrees multiply.

3 Algebraic Closure: Existence and Uniqueness

Definition 3.1 (Algebraically closed; algebraic closure). L is **algebraically closed** if every non-constant $f \in L[x]$ has a root in L (equivalently: splits into linear factors; equivalently: L has no proper algebraic extension). An **algebraic closure** of K is a field $\bar{K}$ that is algebraically closed and algebraic over K .

Theorem 3.2 (Existence of algebraic closures (Artin)). *Every field K has an algebraic closure.*

Proof. Step 1 (one root for everything at once). Let S be the set of monic irreducible polynomials of $K[x]$ and $R = K[x_f : f \in S]$ the polynomial ring in one variable per f . Let $I = (f(x_f) : f \in S) \subseteq R$. Then $1 \notin I$: any relation $1 = \sum_{i=1}^r g_i f_i(x_{f_i})$ lives in the subring $K[x_{f_1}, \dots, x_{f_r}]$, and the quotient of that subring by $(f_1(x_1), \dots, f_r(x_r))$ is $K[x]/(f_1) \otimes_K \dots \otimes_K K[x]/(f_r)$, a tensor product of fields, hence nonzero (S2), so 1 cannot lie in the ideal. By Krull's theorem (S1, via Zorn) extend I to a maximal ideal M ; then $L_1 := R/M$ is a field containing K in which every $f \in S$ — hence every non-constant $f \in K[x]$ — has a root (the image of x_f).

Step 2 (iterate). Apply Step 1 repeatedly: $K \subseteq L_1 \subseteq L_2 \subseteq \dots$ with every non-constant polynomial over L_i having a root in L_{i+1} . Set $L = \bigcup_i L_i$ (a field: operations land in some stage). Every non-constant $f \in L[x]$ has coefficients in some L_i , hence a root in $L_{i+1} \subseteq L$; a field in which every non-constant polynomial has a root is algebraically closed (factor out the root and induct on degree).

Step 3 (cut to algebraic part). Let $\bar{K} = \{\alpha \in L : \alpha \text{ algebraic over } K\}$; this is a field by Corollary 2.8, algebraic over K by construction, and algebraically closed: any non-constant $f \in \bar{K}[x]$ has a root $\beta \in L$; β is algebraic over $\bar{K}$, hence over K (tower of algebraic extensions, Proposition 2.7), so $\beta \in \bar{K}$. Thus $\bar{K}$ is an algebraic closure of K . $\square$

Theorem 3.3 (Uniqueness of algebraic closures). *Any two algebraic closures of K are isomorphic over K (by an isomorphism fixing K pointwise).*

Proof idea. Consider the set of pairs (M, φ) with $K \subseteq M \subseteq \bar{K}_1$ an intermediate field and $\varphi : M \rightarrow \bar{K}_2$ a K -embedding; order by extension. Zorn (S1) gives a maximal (M, φ) . If $M \neq \bar{K}_1$, pick $\alpha \in \bar{K}_1 \setminus M$ with minimal polynomial m over M ; $\varphi(m)$ has a root β in $\bar{K}_2$ (algebraically closed), and φ extends to $M(\alpha) \cong M[x]/(m) \rightarrow \bar{K}_2$, $x \mapsto \beta$ — contradicting maximality. Hence $M = \bar{K}_1$; then $\varphi(\bar{K}_1)$ is algebraically closed inside $\bar{K}_2$ and contains K , so equals $\bar{K}_2$ (algebraicity), i.e. φ is an isomorphism. $\square$

Example 3.4 ($\bar{\mathbb{Q}}$ versus $\mathbb{C}$). • $\bar{\mathbb{Q}}$ is **countable**: it is the union over all $f \in \mathbb{Q}[x]$ (countably many) of the finite root sets of f .

• $\mathbb{C}$ is **uncountable** (Cantor's diagonal); hence $\bar{\mathbb{Q}} \subsetneq \mathbb{C}$, and $e, \pi \in \mathbb{C} \setminus \bar{\mathbb{Q}}$ are transcendental.

- $\bar{\mathbb{Q}}$ is not complete for $|\cdot|$ (its completion is $\mathbb{C}$? no — the completion of $\bar{\mathbb{Q}}$ for $|\cdot|$ is $\mathbb{C}$; the completion of $\mathbb{Q}$ is $\mathbb{R}$, P2); for each p , the completion of $\bar{\mathbb{Q}}$ for $|\cdot|_p$ is $\mathbb{C}_p$ (S18–S19). Closures add *algebraic* points; completions add *limit* points — the two machines of the course, side by side.

4 Preview: Galois Groups and the Horizon $G_{\mathbb{Q}}$

Definition 4.1 (Galois group of an extension, preview). For L/K , write $\text{Aut}(L/K)$ for the group of field automorphisms of L fixing K pointwise. When L/K is finite and *Galois* (separable + normal, S12), this group has order $[L : K]$ and controls all intermediate fields (FTGT, S13).

Example 4.2 (First computations). • $\text{Aut}(\mathbb{Q}(\sqrt{2})/\mathbb{Q}) = \{\text{id}, \sqrt{2} \mapsto -\sqrt{2}\} \cong \mathbb{Z}/2$: the two roots of $x^2 - 2$ are the two “positions” of the point $\sqrt{2}$, and the group permutes them.

- $\text{Aut}(\mathbb{C}/\mathbb{R}) = \{\text{id}, \text{conjugation}\} \cong \mathbb{Z}/2$.
- $\text{Aut}(\mathbb{F}_{p^n}/\mathbb{F}_p)$ is generated by the Frobenius $x \mapsto x^p$ (S12).

Definition 4.3 (The absolute Galois group). $G_K := \text{Gal}(\bar{K}/K) := \text{Aut}(\bar{K}/K)$, equipped with the Krull/profinite topology as an inverse limit of the finite groups $\text{Gal}(L/K)$ over finite Galois subextensions L/K (S16, S34). For $K = \mathbb{Q}$ this is $G_{\mathbb{Q}}$ — the symmetry group of *all* algebraic points of $\mathbb{Q}$.

Remark 4.4 (Why $G_{\mathbb{Q}}$ is the horizon of the course). Class field theory (Act VI) computes the abelianization $G_{\mathbb{Q}}^{\text{ab}}$ completely (Kronecker–Weber, S16/S47–S49); the étale fundamental group (S35) identifies G_K with $\pi_1^{\text{ét}}(\text{Spec } K)$, making Galois theory a special case of topology-of-schemes; Galois cohomology (S45) measures its failure to be trivial; and Deligne’s theorem (S58) is the logical shadow of “enough points”. Everything grows from the single object $\bar{K}$ built in Theorem 3.2.

5 Exercises

Exercise 1. Prove Proposition 1.4 and Proposition 1.5 in full detail, and identify the prime subfield of $\mathbb{F}_{p^n}$, $\mathbb{Q}_p$, and $\mathbb{C}$.

Exercise 2. Fill in the abelian-group lemma used in Proposition 1.7: if $\text{ord}(a) = r$ and $\text{ord}(b) = s$ in an abelian group, some element has order $\text{lcm}(r, s)$. Conclude that $\mathbb{F}_q^{\times}$ is cyclic.

Exercise 3. Using the tower law (Proposition 2.3), compute $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}]$ by exhibiting an explicit basis; then verify $\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2} + \sqrt{3})$ as in Example 2.6.

Exercise 4. Find the minimal polynomials over $\mathbb{Q}$ of: $\sqrt{3} + \sqrt{5}$; $\sqrt[3]{2} + 1$; $i\sqrt{2}$; and $\cos(2\pi/5)$ (hint: $\zeta_5 + \zeta_5^{-1}$; cyclotomics appear in S9).

Exercise 5. Show that $\bar{\mathbb{Q}}/\mathbb{Q}$ is algebraic but $[\bar{\mathbb{Q}} : \mathbb{Q}] = \infty$ (consider $\sqrt[n]{2}$ for all n). Conclude that the converse of Proposition 2.7(a) fails.

Exercise 6. Prove that for algebraic α over K , $K(\alpha) = K[\alpha]$ (every element is a polynomial in α), and compute the inverse of $1 + \sqrt{2} + \sqrt{3}$ inside $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ explicitly.

Exercise 7. In Step 1 of Theorem 3.2, prove carefully that $K[x]/(f_1) \otimes_K \cdots \otimes_K K[x]/(f_r) \neq 0$ using only S2 (tensor products of free modules, base change) — this is exactly where the linear layer of S2 earns its keep.

Exercise 8. Show that a field L in which every non-constant polynomial has a root is algebraically closed (induction on degree, factoring out roots).

Exercise 9. Prove that $\mathbb{Q}$ is countable while $\mathbb{C}$ is not; deduce the existence of transcendentals without naming any. Then show $\mathbb{Q}(\pi) \cong \mathbb{Q}(x)$ (rational function field).

Exercise 10. Compute $\text{Aut}(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q})$ explicitly (four automorphisms; identify the group as $\mathbb{Z}/2 \times \mathbb{Z}/2$) and match each automorphism to a choice of signs for $(\sqrt{2}, \sqrt{3})$ — a first taste of the FTGT dictionary (S13).

Exercise 11. Show that any finite subgroup of $\text{Aut}(L/K)$ has order at most $[L : K]$ (hint: linear independence of characters — Dedekind’s lemma, proved in S12; accept it here as a black box if you prefer).

Where This Session Feeds the Course

- **Depends on:** S1 (PID structure of $K[x]$, Krull/Zorn), S2 (tensor products in Step 1 of Theorem 3.2), P1 (Lagrange, cyclic groups).
- **Feeds:** S12–S16 (Galois theory, finite fields, $G_{\mathbb{Q}}$); S20 (degrees e, f of valuation extensions); S38 (Riemann surfaces over $\mathbb{C}$); S45 (Galois cohomology of G_K); S50 (Lubin–Tate over $\mathbb{Q}_p$).
- **Unifying thread:** a field is an arena of points (roots); an extension adds points; the minimal polynomial is the equation of a point; the algebraic closure $\bar{K}$ is the space of *all* algebraic points, and $G_K = \text{Aut}(\bar{K}/K)$ is the group permuting them — evaluation and symmetry, unified.

References for this session

- [1] Lang, S. *Algebra*, ch.V (extensions, algebraic closures).
- [2] Artin, E. *Galois Theory*, ch.1–3 (the classical route, incl. Artin’s proof).
- [3] Conrad, K. *Fields and Galois Theory* notes (free, detailed proofs).
- [4] Stewart, I. *Galois Theory*, ch.1–5 (gentle companion).

End of Session S3 — next: Session S4, Category Theory Essentials